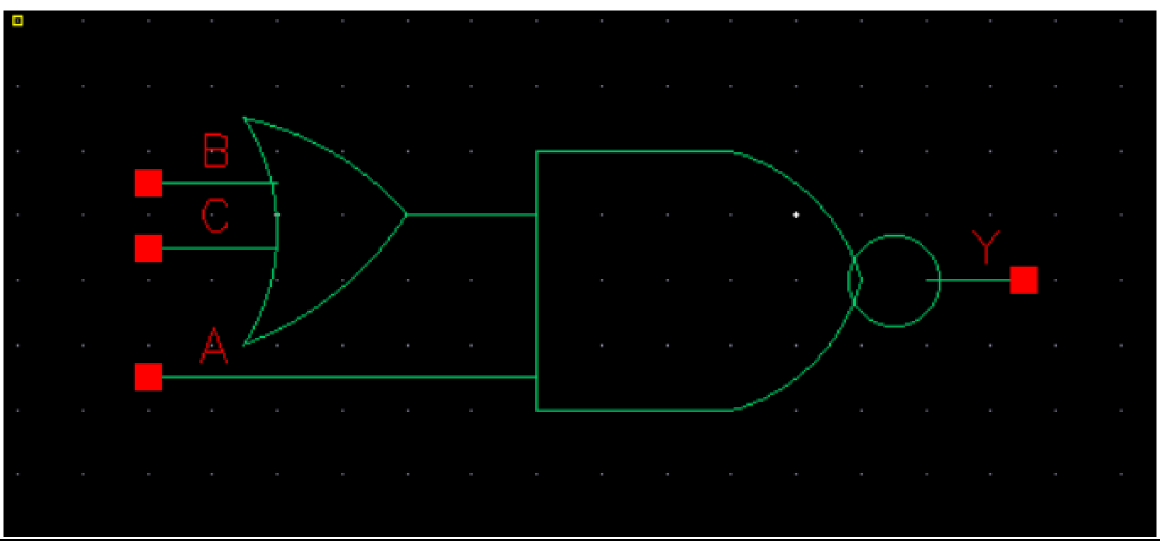
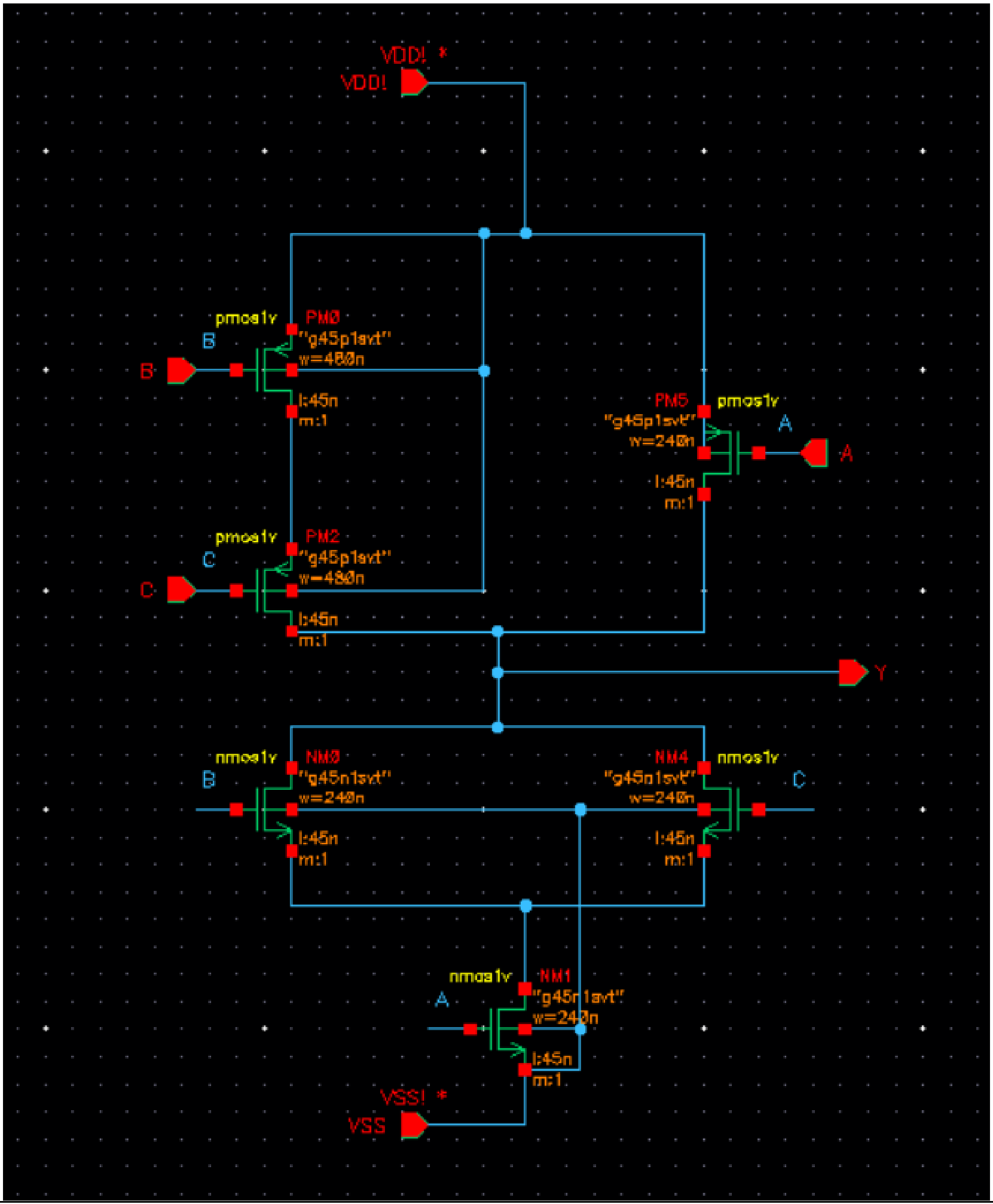
Library Name:	**cgm**		
Cell Name:	**AOI3**		
Function/Truth Table: Y = $\overline{(\boldsymbol{B}+\boldsymbol{C}).\boldsymbol{A}}$			
A	**B**	**C**	**Y**
0	**0**	**0**	**1**
0	**0**	**1**	**1**
0	**1**	**0**	**1**
0	**1**	**1**	**1**
1	**0**	**0**	**1**
1	**0**	**1**	**0**
1	**1**	**0**	**0**
1	**1**	**1**	**0**
Propagation Delay (path based):			
Transition			**Delay (s)**
A → **Y** ↑ (T_PLH)			**1.25n**
A → **Y** ↓ (T_PHL)			**1.326n**
B → **Y** ↑ (T_PLH)			**1.397n**
B → **Y** ↓ (T_PHL)			**1.633n**
C → Y ↑ (T_PLH)			**1.369n**
C → Y ↓ (T_PHL)			**1.682n**

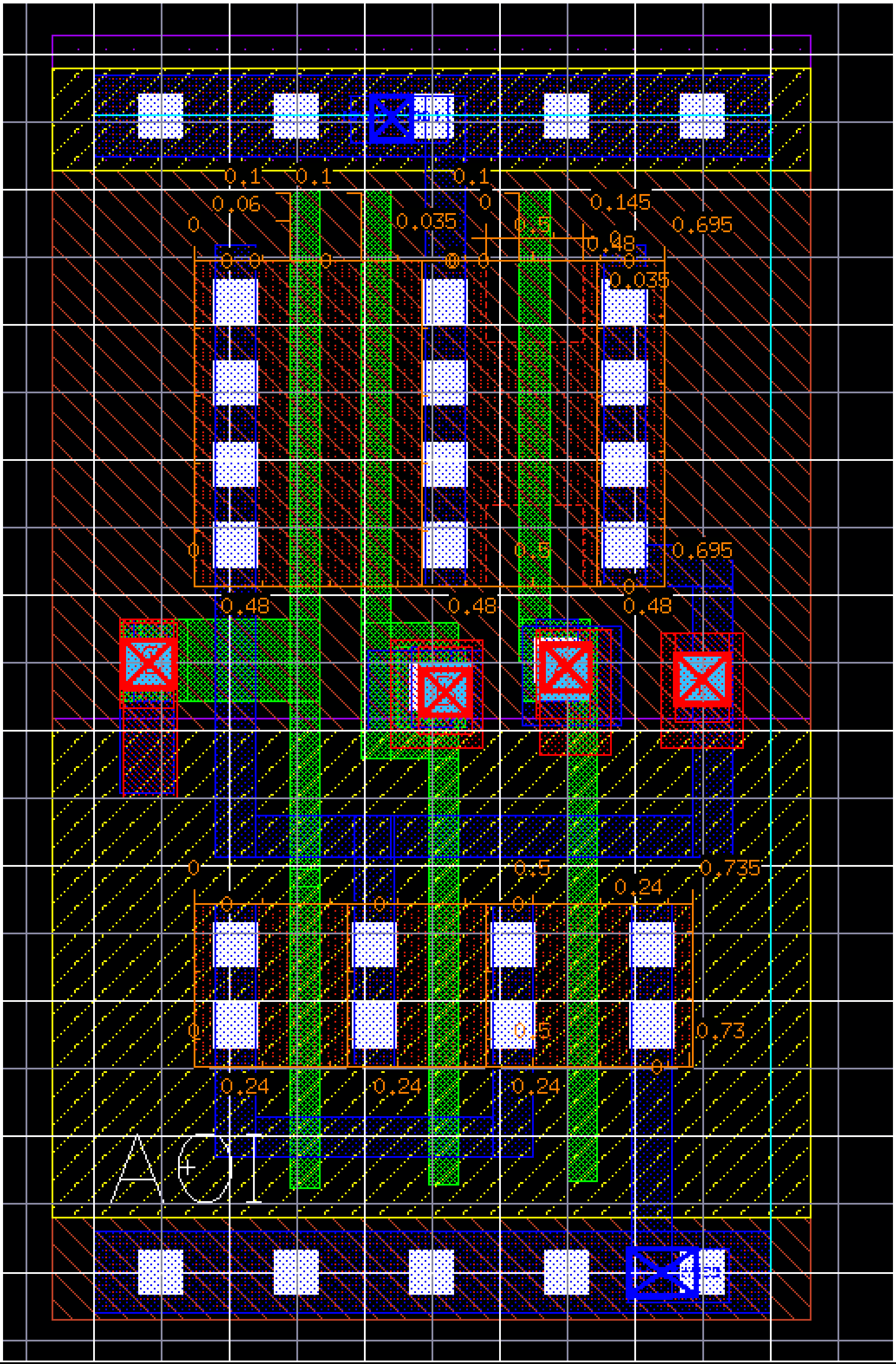
Output Rise Time (path based):	
Transition	**Rise Time (s)**
A → Y ↑	**2.484n**
B → Y ↑	**2.679n**
C → Y ↑	**2.652n**
Output Fall Time (path based):	
Transition	**Fall Time (s)**
A → Y ↓	**2.457n**
B → Y ↓	**3.104n**
C → Y ↓	**3.234n**
Layout Height: 1.71µm	
Layout Width: 1µm	
Layout Area: 1.71µm^2	
Symbol with Port Names:	



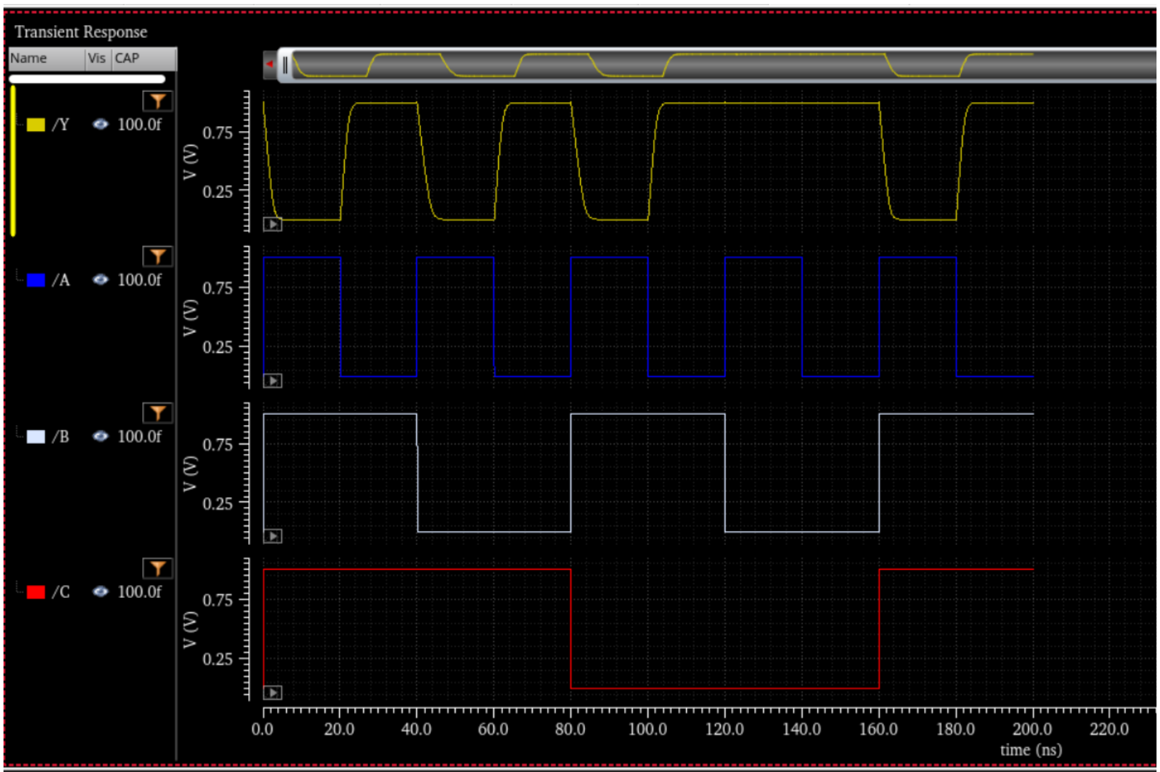
Schematic:


VDD! *
VDD!
pmos1v
PM0
B
"g45p1svt"
w=480n
l:45n
m:1
B
pmos1v
PM2
C
"g45p1svt"
w=480n
l:45n
m:1
C
PM5
pmos1v
"g45p1svt"
w=240n
l:45n
m:1
A
A
Y
nmos1v
NM0
B
"g45n1svt"
w=240n
l:45n
m:1
NM4
nmos1v
"g45n1svt"
w=240n
l:45n
m:1
C
nmos1v
NM1
A
"g45n1svt"
w=240n
l:45n
m:1
VSS! *
VSS

Layout:


0.1
0.1
0.1
0.06
0.035
0.145
0.695
0.48
0.035
0.5
0.695
0.48
0.48
0.48
0.5
0.735
0.24
0.5
0.73
0.24
0.24
0.24
AOI

Functional Simulation Waveforms



Comments/Notes:

The AOI3 (AND-OR-INVERT) gate was successfully tested, confirming correct functional verification. The extracted parasitics have impacted timing, with rise times increasing compared to the pre-layout results, while fall times have improved. This behavior is due to the influence of parasitic capacitances and resistances in the extracted layout, which affect the charge and discharge paths differently. Despite these variations, the gate meets expected performance requirements and remains functionally accurate.